Amazon Sales Database Management System

1. Abstract

This project focuses on designing and managing an **Amazon Sales Database** using MySQL. The system stores and analyzes sales transactions including customer details, product information, payment methods, and ratings. The main objective is to demonstrate database creation, data manipulation, data analysis, and relational operations using SQL queries.

The project implements various SQL concepts such as table creation, data insertion, aggregate functions, string functions, date functions, joins, subqueries, window functions, and transaction control commands. Through this database, sales performance, customer behavior, and product category analysis can be efficiently performed.

This project helps in understanding how real-world e-commerce platforms manage and analyze large volumes of transactional data.

```sql
create database amazon_sales;
use amazon_sales;
CREATE TABLE sales (
    order_id INT PRIMARY KEY,
    order_date DATE,
    customer_name VARCHAR(100),
    city VARCHAR(50),
    product_category VARCHAR(50),
    product_name VARCHAR(100),
    quantity INT,
    price DECIMAL(10,2),
    payment_method VARCHAR(50),
    rating DECIMAL(2,1)
);
INSERT INTO sales VALUES
(1,'2024-01-01','Arun','Chennai','Electronics','Headphones',2,1500,'UPI',4.5),
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	order_date	customer_name	city	product_category	product_name	quantity	price	payment_method	rating	customer_id
1	2024-01-01	Arun	Chennai	Electronics	Headphones	2	1500.00	UPI	4.5	101
2	2024-01-02	Priya	Mumbai	Clothing	T-Shirt	3	800.00	Card	4.0	102
3	2024-01-03	Rahul	Delhi	Electronics	Keyboard	1	1200.00	Cash	3.8	103
4	2024-01-05	Sneha	Chennai	Home	Mixer	1	2500.00	UPI	4.7	104
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid

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2. Introduction

In modern e-commerce platforms like Amazon, large amounts of sales data are generated daily. Managing this data efficiently requires a structured database system.

MySQL is a widely used Relational Database Management System (RDBMS) that allows storing, retrieving, and analyzing structured data efficiently.

This project simulates a small-scale Amazon sales system where:

- Customer orders are stored
- Product sales are tracked
- Payment methods are recorded
- Customer memberships are maintained
- Sales analytics are performed

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3. Objectives

The main objectives of this project are:

- To create and manage a relational database using MySQL.
- To perform CRUD operations (Create, Read, Update, Delete).
- To apply aggregate and analytical SQL functions.
- To understand joins and relationships between tables.
- To analyze sales data for business insights.
- To implement transaction management.

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4. Tools and Technologies Used

Component	Technology
Database	MySQL
Query Language	SQL
Platform	MySQL Workbench / Command Line
Database Type	Relational Database

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5. Database Design

5.1 Database Creation

The database named:

```
amazon_sales
```

is created to store all sales-related information.

5.2 Tables Used

Table 1: sales

Stores order and product transaction details.

Column Name	Data Type	Description
order_id	INT	Unique order identifier
order_date	DATE	Date of purchase
customer_name	VARCHAR	Customer name
city	VARCHAR	Customer city
product_category	VARCHAR	Category of product
product_name	VARCHAR	Product purchased
quantity	INT	Quantity ordered
price	DECIMAL	Product price
payment_method	VARCHAR	Mode of payment
rating	DECIMAL	Customer rating
customer_id	INT	Foreign key reference

Table 2: customers

Stores customer membership information.

Column Name	Data Type	Description
customer_id	INT	Customer ID
customer_name	VARCHAR	Name
city	VARCHAR	Location
membership	VARCHAR	Prime/Regular

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6. SQL Operations Performed

6.1 Data Definition Language (DDL)

- CREATE DATABASE
- CREATE TABLE
- ALTER TABLE

Example:

```
CREATE TABLE sales (...);
```

6.2 Data Manipulation Language (DML)

- INSERT records
- UPDATE records
- DELETE records

Example:

```
INSERT INTO sales VALUES (...);
```

6.3 Data Retrieval (SELECT Queries)

Basic data retrieval:

```
SELECT * FROM sales;
```

6.4 Calculated Columns

Total order amount calculation:

```
quantity * price AS total_amount
```

Used to determine revenue per order.

6.5 Aggregate Functions

Used for business analysis:

- COUNT() → total orders

- SUM() → total sales
- AVG() → average rating
- MAX() / MIN() → price analysis

6.6 GROUP BY and HAVING

Category-wise sales analysis:

```
GROUP BY product_category
HAVING total_sales > 2000;
```

Helps identify high-performing categories.

6.7 String Functions

- UPPER()
- LOWER()
- CONCAT()
- LENGTH()

Used for formatting and text analysis.

6.8 Date Functions

Extracted components from order date:

- YEAR()
- MONTH()
- DAY()

Useful for time-based sales reporting.

6.9 Conditional Function

```
IF(rating >= 4,'Good','Average')
```

Classifies customer reviews.

6.10 Sorting

Orders arranged by price:

```
ORDER BY price DESC;
```

6.11 Subqueries

Customers paying above average price:

```
WHERE price > (SELECT AVG(price) FROM sales);
```

6.12 Window Functions

Ranking orders:

- ROW_NUMBER()
- RANK()

Used for advanced analytics.

6.13 Joins

INNER JOIN

Shows matching customer records.

LEFT JOIN

Shows all sales even without customer data.

RIGHT JOIN

Shows all customers including non-buyers.

UNION

Combines LEFT and RIGHT join results.

6.14 Transaction Control Language (TCL)

- DELETE
- ROLLBACK
- COMMIT

Ensures data safety and recovery.

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7. Results and Analysis

The database enables:

- Identification of best-selling product categories.
- Calculation of total revenue.
- Customer performance analysis using ratings.
- Membership-based customer tracking.
- Ranking products based on price.

Key Findings:

- Electronics category generated higher revenue.
- Customers with ratings ≥ 4 are categorized as good reviewers.
- Window functions help rank sales efficiently.

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8. Conclusion

The Amazon Sales Database project successfully demonstrates the use of MySQL in managing and analyzing e-commerce data. Various SQL concepts including joins, aggregate functions, window functions, and transactions were implemented effectively. The project provides practical knowledge of database operations and simulates real-world business analytics scenarios.